

Mid Exam Question Key and Grading Scheme

Grading Scheme

Q. No.	Part	Marks	CO	What to award
Q1	Definition of AI	1	CO1	Correct definition of AI
Q1	Narrow AI vs Generative AI	2	CO1	At least two clear differences
Q1	Civil engineering examples	2	CO1	1 mark for each correct example
Q2	(a) AI-ML-DL diagram	2	CO1	Correct nested relationship: AI $\supset$ ML $\supset$ DL
Q2	(b) DL vs traditional ML	2	CO1	Any two valid differences
Q2	(c) Concrete strength prediction	1	CO1	Correct link to concrete strength
Q3	(a) Interpret $r = 0.78$	2	CO2	Positive, strong/moderate-strong relationship
Q3	(b) Causation	2	CO2	No; correlation $\neq$ causation; explain confounding/reverse
Q3	(c) Confounder	1	CO2	Any one valid variable
Q4	Visualization techniques	5	CO3	1 mark each: description + when used
Q5	(a) Mean, median, better measure	2	CO4	Correct mean, median, justified choice
Q5	(b) IQR and outliers	2	CO4	Q1, Q3, IQR, fences, outliers
Q5	(c) Remove outlier?	1	CO4	One reason for and one against
Q5 OR	(a) Learning paradigm	2	CO4	Supervised + justification
Q5 OR	(b) Bias sources	2	CO4	Any two valid sources
Q5 OR	(c) Explainable AI	1	CO4	Correct concept + example

Total: 25 marks

Q1. Define Artificial Intelligence and explain Narrow AI vs Generative AI. Give one civil engineering example for each.

Artificial Intelligence: AI is the branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, perception, decision-making, and problem-solving.

Feature	Narrow AI	Generative AI
Scope	Performs one specific task	Creates new content/data
Intelligence	Limited to a narrow domain	Learns patterns and generates new outputs
Example	Crack detection from concrete images	Generating concrete mix design alternatives or synthetic bridge images

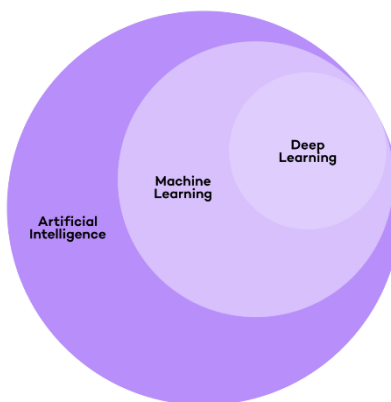
Civil engineering examples:

- **Narrow AI:** AI model that detects cracks in concrete bridge images.
- **Generative AI:** AI that generates alternative structural layouts or synthetic site images for planning.

Q2. AI, ML, DL relationship; DL vs traditional ML; concrete strength prediction

(a) Diagram:

Artificial Intelligence
└── Machine Learning
 └── Deep Learning



AI is the broad field. ML is a subset where machines learn from data. DL is a subset of ML using deep neural networks.

(b) Deep Learning vs Traditional Machine Learning:

Traditional ML	Deep Learning
Requires manual feature engineering	Learns features automatically
Works with smaller datasets	Usually needs large datasets
Often uses algorithms like SVM, regression, random forest	Uses neural networks, CNNs, RNNs, transformers
More interpretable	Often less interpretable

(c) Concrete strength prediction: Traditional ML can predict concrete compressive strength using input features such as cement, water, aggregate, age, and curing. Deep Learning can automatically learn complex patterns from larger datasets, sensor data, or images of concrete microstructure.

Q3. Correlation and limitations; $r = 0.78$ between humidity and rainfall

Correlation: A statistical measure of the strength and direction of association between two variables. Pearson's r ranges from -1 to $+1$. **Limitations:** Shows only linear relationship, sensitive to outliers, does not prove causation, may be affected by confounding variables.

(a) $r = 0.78$ means a **strong positive relationship**. As afternoon humidity increases, rainfall tends to increase.

(b) No. Correlation does not prove causation. High humidity may be associated with rainfall, but it does not necessarily cause rainfall. There may be confounding variables or reverse causality.

(c) One possible confounder: **monsoon season** or **temperature**.

Q4. Visualization techniques and when to use them

Technique	Use when
Scatter plot	To show relationship between two continuous variables
Histogram	To show distribution/frequency of one continuous variable
Boxplot	To show median, quartiles, spread, and outliers; compare groups
Bar plot	To compare categorical values or group totals
Correlation matrix	To show pairwise correlations among many numerical variables

Q5. Temperature data: mean, median, IQR, outliers

Give data: **22, 24, 30, 31, 48, 32, 31, 29, 28, 24, 23, 21**

Mean and median:

Mean = $343 / 12 = 28.58^{\circ}\text{C}$

Sorted data: **21, 22, 23, 24, 24, 28, 29, 30, 31, 31, 32, 48**

Median = $(28 + 29) / 2 = 28.5^{\circ}\text{C}$

IQR and outliers:

$Q1 = \text{median}(21, 22, 23, 24, 24, 28) = 23.5$

$Q3 = \text{median}(29, 30, 31, 31, 32, 48) = 31.0$

$IQR = Q3 - Q1 = 31.0 - 23.5 = 7.5$

Lower fence = $Q1 - 1.5 * IQR = 23.5 - 1.5 * 7.5 = 12.25$

Upper fence = $Q3 + 1.5 * IQR = 31.0 + 1.5 * 7.5 = 42.25$

Outliers: values below **12.25** or above **42.25**. So **48 is an outlier**.

(c) Should outlier be removed?

- **For:** If it is due to data entry error or instrument fault, removing it improves accuracy.
 - **Against:** If it is a genuine extreme event, removing it hides important climate variability.
-

Q5 OR. AI for automated quality inspection of concrete surfaces using drone images

(a) Learning paradigm: Supervised learning is most suitable. Drone images can be labelled as “crack”, “spalling”, “honeycombing”, or “no defect”. The model learns from these labelled examples to classify or detect defects. Unsupervised learning may help if labels are unavailable, but supervised learning is best for known defect categories.

(b) Two potential sources of bias:

1. Training images collected only in good daylight or one weather condition; model may fail under shadows/glare.
2. Training data from only one concrete type or surface finish; model may not generalise to other sites.

(c) Explainable AI: Explainable AI makes model decisions understandable. For example, Grad-CAM or saliency maps can highlight the image region that caused the model to predict a crack. This helps engineers verify the result, trust the system, and identify bias or errors.